

Sam Gunn

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POSITIONS	Massachusetts Institute of Technology C.L.E. Moore Instructor of Mathematics (Summer 2026 to Present)	
EDUCATION	The University of California, Berkeley Ph.D. in Computer Science (Fall 2020 to Summer 2026) Advisor: Umesh Vazirani The University of Texas at Austin B.S. in Mathematics (Fall 2016 to Spring 2020) Advisors: Scott Aaronson and Cameron Gordon	
RESEARCH	<i>Journal publications</i> 1. Omar Alrabiah, Prabhanjan Ananth, Miranda Christ, Yevgeniy Dodis, and Sam Gunn. “Ideal pseudorandom codes.” <i>SIAM Journal on Computing</i>, Special Issue for the 57th Annual ACM Symposium on Theory of Computing (STOC). ePrint: 2024/1840. 2. Sam Gunn, Nathan Ju, Fermi Ma, and Mark Zhandry. “Commitments to quantum states.” <i>SIAM Journal on Computing</i>, Special Issue for the 55th Annual ACM Symposium on Theory of Computing (STOC). ePrint: 2022/1358. 3. Scott Aaronson and Sam Gunn. “On the classical hardness of spoofing linear cross-entropy benchmarking.” <i>Theory of Computing</i> 16(11):1–8 (2020). DOI: 10.4086/toc.2020.v016a011. arXiv: 1910.12085. <i>Conference publications</i> 1. Miranda Christ, Noah Golowich, Sam Gunn, Ankur Moitra, and Daniel Wichs. “Improved pseudorandom codes from permuted puzzles.” <i>Proceedings of the 58th Annual ACM Symposium on Theory of Computing (STOC)</i>. 2026. DOI: 10.1145/3798129.3800916. ePrint: 2025/2222. 2. Sanjam Garg, Sam Gunn, and Mingyuan Wang. “Black-box crypto is useless for pseudorandom codes.” <i>Proceedings of the 23rd Theory of Cryptography Conference (TCC)</i>. 2025. DOI: 10.1007/978-3-032-12290-2_7. ePrint: 2025/1021. Outstanding Paper Award. 3. Sam Gunn, Xuandong Zhao, and Dawn Song. “An undetectable watermark for generative image models.” <i>The Thirteenth International Conference on Learning Representations (ICLR)</i>. 2025. arXiv: 2410.07369. 4. Xuandong Zhao, Sam Gunn, Miranda Christ, Jaiden Fairoze, Andres Fabrega, Nicholas Carlini, Sanjam Garg, Sanghyun Hong, Milad Nasr, Florian Tramer, Somesh Jha, Lei Li, Yu-Xiang Wang, and Dawn Song. “SoK: Watermarking for AI-generated content.” <i>Proceedings of the 46th IEEE Symposium on Security and Privacy (S&P)</i>. 2025. DOI: 10.1109/SP61157.2025.00178. arXiv: 2411.18479. 5. Omar Alrabiah, Prabhanjan Ananth, Miranda Christ, Yevgeniy Dodis, and Sam Gunn. “Ideal pseudorandom codes.” <i>Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)</i>. 2025. DOI: 10.1145/3717823.3718309. ePrint: 2024/1840. 6. Sam Gunn and Ramis Movassagh. “Quantum one-time protection of any randomized algorithm.” <i>Proceedings of the 45th Annual International Cryptology Conference (CRYPTO)</i>. 2025. DOI: 10.1007/978-3-032-01878-6_11. ePrint: 2024/1798. 7. Sam Gunn, Yael Tauman Kalai, Anand Natarajan, and Ági Villányi. “Classical commitments to quantum states.” <i>Proceedings of the 57th Annual ACM Symposium</i>	

- on Theory of Computing (STOC). 2025. DOI: [10.1145/3717823.3718264](#). ePrint: [2024/606](#). Presented at *Quantum Information Processing (QIP)*.
8. Miranda Christ and Sam Gunn. “Pseudorandom error-correcting codes.” *Proceedings of the 44th Annual International Cryptology Conference (CRYPTO)*. 2024. DOI: [10.1007/978-3-031-68391-6_10](#). ePrint: [2024/235](#).
 9. Miranda Christ, Sam Gunn, and Or Zamir. “Undetectable watermarks for language models.” *Proceedings of the 37th Annual Conference on Learning Theory (COLT)*. 2024. ePrint: [2023/763](#). Presented at *Real World Crypto (RWC)*.
 10. Andrea Coladangelo and Sam Gunn. “How to use quantum indistinguishability obfuscation.” *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: [10.1145/3618260.3649779](#). ePrint: [2023/1756](#).
 11. Thiago Bergamaschi, Louis Golowich, and Sam Gunn. “Approaching the quantum Singleton bound with approximate error correction.” *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*. 2024. DOI: [10.1145/3618260.3649680](#). arXiv: [2212.09935](#). Presented at *Quantum Information Processing (QIP)*.
 12. Sam Gunn, Nathan Ju, Fermi Ma, and Mark Zhandry. “Commitments to quantum states.” *Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC)*. 2023. DOI: [10.1145/3564246.3585198](#). ePrint: [2022/1358](#). Presented at *Quantum Information Processing (QIP)*.
 13. Scott Aaronson and Sam Gunn. “On the classical hardness of spoofing linear cross-entropy benchmarking.” 2020. arXiv: [1910.12085](#). Presented at the *Twenty-Second Annual Southwest Quantum Information and Technology Workshop (SQuInT)*.

Manuscripts

1. Sam Gunn. “How to sketch a learning algorithm.” 2026. arXiv: [2604.07328](#). In submission.

AWARDS

- Google PhD Fellowship 2022–2025
- Lotfi A. Zadeh Prize 2024–2025

TALKS

- “How to Sketch a Learning Algorithm”
- MIT Statistics and Data Science Seminar September 2026
 - MIT ML+Cryptography Seminar May 2026
 - Institute for Advanced Study May 2026
 - Flatiron Institute June 2026
 - Theory for Trustworthy and Interpretable AI at STOC 2026 June 2026
 - Invited talk at ITC 2026 August 2026
- “On Undetectable Backdoors”
- UT Austin Theory and Alignment Group November 2025
- “Pseudorandom Error-Correcting Codes”
- NTT Research March 2024
 - Google Research NYC April 2024
 - CRYPTO 2024 August 2024
 - NYU Cryptography Seminar March 2025
 - Harvard Theory Seminar September 2025
 - TCC 2025 December 2025
- “Undetectable Watermarks for Language Models”
- Google Research Mountain View July 2023
 - MIT Cryptography Group August 2023
 - NYU Cryptography Seminar ([video](#)) September 2023

	• Berkeley ML Theory Seminar	September 2023
	“Quantum One-Time Protection of Any Randomized Algorithm”	
	• CRYPTO 2025	August 2025
	“Quantum Indistinguishability Obfuscation”	
	• CIQC Spark Talk	March 2024
	• University of Ottawa Quantum Seminar	April 2024
	“Approaching the Quantum Singleton Bound with Approximate Error Correction”	
	• Simons Quantum Colloquium (video)	December 2022
	“Commitments to Quantum States”	
	• STOC 2023 (video)	June 2023
	“Unclonable Cryptography”	
	• Guest lecture for Quantum Complexity, UC Berkeley	December 2022
	“Certified Entropy from Random Circuit Sampling”	
	• Berkeley Theory Lunch	March 2022
SERVICE	Program Committee, ITCS	2027
	MIT Computer Science UROP Mentor	Summer 2026
	Berkeley Computer Science DRP Mentor	Spring 2021
REVIEWING	STOC, FOCS, SODA, CRYPTO, EUROCRYPT, TCC, NeurIPS, ICLR, QIP, Annals of Statistics	